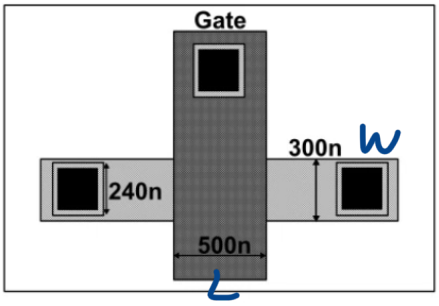
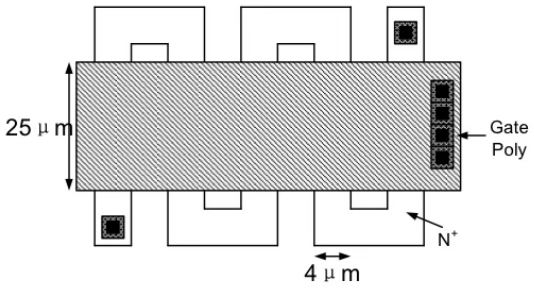
Exercise 1

1-1. Two layouts of n-channel MOSFET are shown in Fig.1.1. What is the width and length of the two devices?



(a)



(b)

Fig.1.1

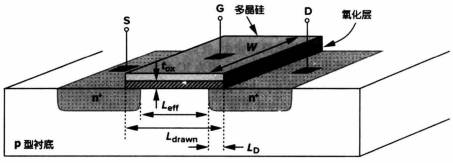


图 2.2 MOS 器件的结构

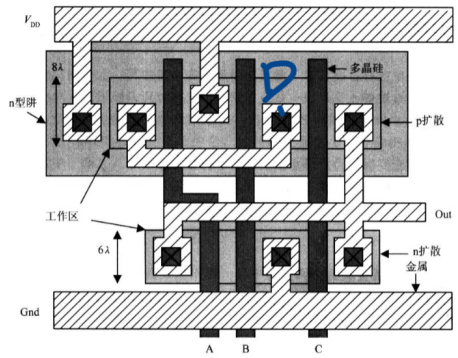
栅沿源漏通道的横向尺寸叫栅长 L，与之垂直方向的栅的尺寸叫做栅宽 W。由于在制造

解. (a) $W=300n$, $L=500n$

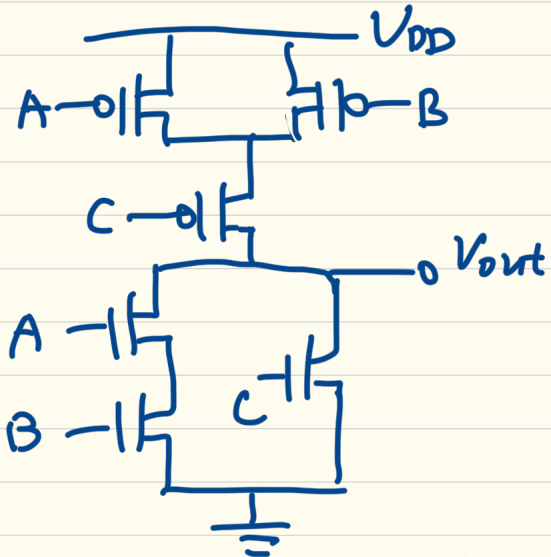
(b) $W=4\mu m$

$L=25\mu m \times 5 = 125\mu m$

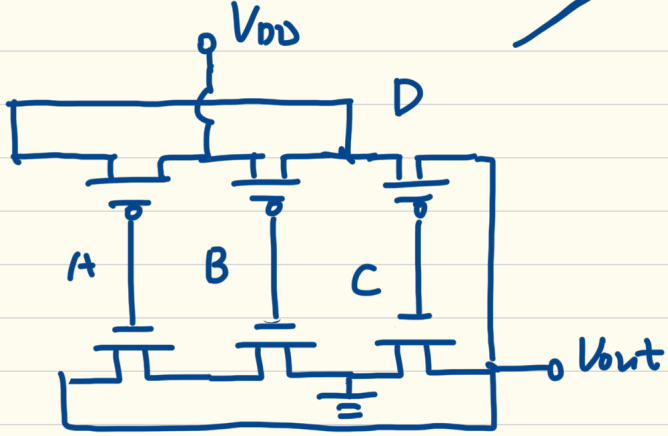
1-2. The layouts of a circuit are shown in Fig.1.2. Give the corresponding schematics and its function, and mark the W/L sizes of each transistor. Assume $L=2\lambda$, $\lambda=0.4\mu m$.



(a) p-sub, n-well technology



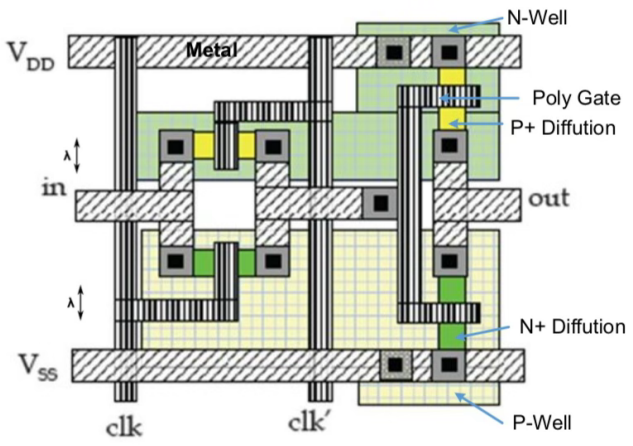
解:



For PMOS, $\frac{W}{L} = \frac{3.2\mu m}{0.3\mu m}$

NMOS, $\frac{W}{L} = \frac{2.4\mu m}{0.3\mu m}$

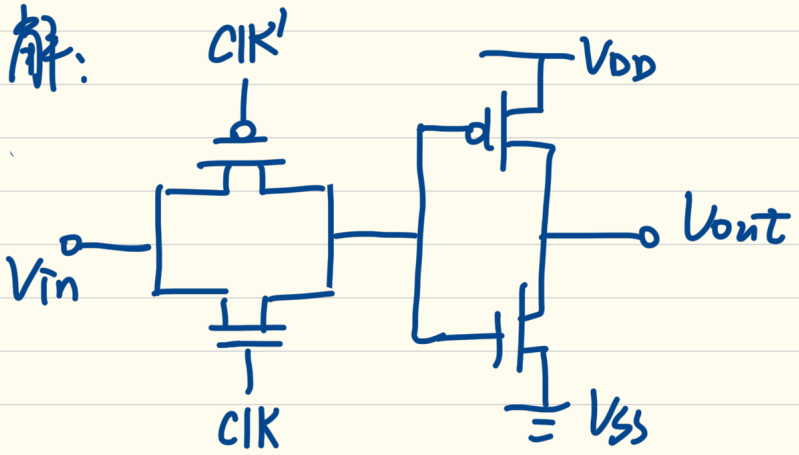
$V_{out} = (AB+C)'$



(b) dual-well technology, and $clk' = \overline{clk}$

Fig.1.2

解:



For All, $\frac{W}{L} = \frac{0.4\mu m}{0.3\mu m}$

CIK=1, Vout= $\overline{Vin}$; CIK=0, Vout=0

1-3. Layout of a different pair with PMOS current source loads in p-sub N-well technology is shown as Fig.1.3. Give the corresponding schematics and mark the W/L sizes of each transistor. Assume λ=0.4μm.

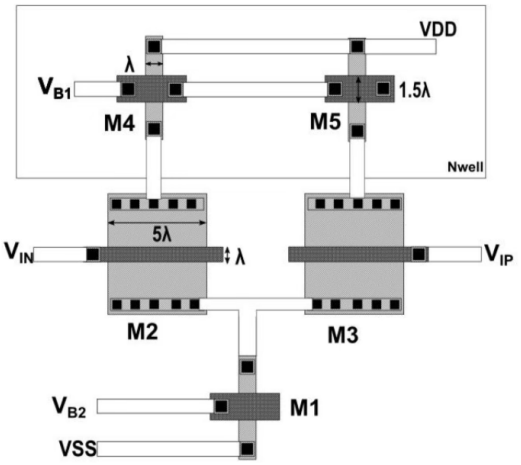


Fig.1.3

解:

